

The Best Open Source Machine Learning Frameworks

In this article, we present what the author rates as the top eight open source machine learning frameworks.



Learning may be defined as the process of improving one's ability to perform a task efficiently. Machine learning is another sub-field of computer science, which enables modern computers to learn without being explicitly programmed. Machine learning has basically evolved from artificial intelligence via pattern recognition and computational learning theory. Machine learning explores the area of algorithms, which can make high end predictions on data. In recent times, machine learning has been deployed in a wide range of computing tasks, where designing efficient algorithms and programs becomes rather difficult, such as email spam filtering, optical character recognition, search engine improvement, digital image processing, data mining, etc.

Tom M. Mitchell, renowned computer scientist and professor at Carnegie Mellon University, USA, defined machine learning as: "A computer program is said to learn from experience E with respect to some class of tasks T and performance measure P , if its performance at tasks in T , as measured by P , improves with experience E ."

Machine learning tasks are broadly classified into three categories, depending on the nature of the learning 'signal' or 'feedback' available to a learning system.

- *Supervised learning* is regarded as a machine learning

task of inferring a function from labelled training data. In supervised learning, each example is a pair consisting of an input object (vector) and a desired output value (supervisory signal).

- *Unsupervised learning*: This is regarded as the machine learning task of inferring a function to describe hidden structures from unlabelled data. It is closely related to the problem of density estimation in statistics.
- *Reinforcement learning* is an area of machine learning that is linked to how software agents take actions in the environment so as to maximise some notion of cumulative reward. It is applied to diverse areas like game theory, information theory, swarm intelligence, statistics and genetic algorithms. In machine learning, the environment is formulated as a Markov decision process (MDP) due to dynamic programming techniques.

The application of machine learning to diverse areas of computing is gaining popularity rapidly, not only because of cheap and powerful hardware, but also because of the increasing availability of free and open source software, which enable machine learning to be implemented easily. Machine learning practitioners and researchers, being a part of the software engineering team, continuously build sophisticated products, integrating intelligent algorithms